



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 07

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Objective of Lab No. 07:

After performing lab 7, students will be able to:

- Create and Open Text Files using Python
- Read and Write Data from and into the files

Lab Tasks:

Lab Task 1: Basic File Writing

Objective: Learn how to create and write data to a file.

Task:

- Create a file named student.txt
- Write the following details into the file:
 - Name
 - Program
 - Semester
- Use the write() method



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- Close the file properly

Lab Task 2: Writing Multiple Lines

Objective: Understand writelines().

Task:

- Create a list of 5 student names
- Write all names into a file students.txt using writelines()
- Each name should appear on a new line

Expected Concepts:

- Lists
- writelines()
- \n newline character

Lab Task 3: Reading File Content

Objective: Learn different reading methods.

Task:

- Open students.txt
- Read:
 1. Entire file using read()
 2. First line using readline()
 3. All lines using readlines()
- Print outputs clearly



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: Count Lines, Words, Characters

Objective: Perform analysis on file content.

Task:

- Read a text file (**Put Some Data Before Manually**)
- Count:
 - Total number of lines
 - Total number of words
 - Total number of characters
- Display results

Lab Task 4: Search in File

Objective: Implement searching.

Task:

- Ask user for a word
- Search that word in a file
- Display:
 - Number of occurrences
 - Line numbers where it appears

Lab Task 5: Copy File Content

Objective: Work with multiple files.

Task:

- Copy content from source.txt to destination.txt



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- Ensure exact duplication

Lab Task 6: Mini File-Based System

Objective: Build a small system.

Task:

Create a menu-driven program:

1. Add student record
2. View all records
3. Search student
4. Exit

Store all data in a file.